

- 4 Adult *Drosophila* fruit flies feed on yeasts on the surface of ripe fruits. Female flies lay their eggs inside the fruit and the larvae feed inside the fruit before developing into adults.

The tree *Morinda citrifolia* produces fruits called noni fruits. Noni fruits contain a toxin and are avoided by nearly all *Drosophila* species, including *Drosophila yakuba* in mainland Africa.

One population of *D. yakuba* lives on the island of Mayotte off the coast of Africa. The population of *D. yakuba* on Mayotte has specialised to feed on noni fruit. Flies of this population produce enzymes that give them resistance to the toxin in noni fruits. Adults are able to feed on yeasts on the fruit surface and their eggs and larvae can develop and grow inside the fruit.

Scientists have used bioinformatics to estimate that the unique *D. yakuba* population on Mayotte originated in the last 30 000 years.

- (a) Describe how a population of *Drosophila* can develop resistance to noni fruit toxin.

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- (b) Explain why living on an island increases the likelihood of the noni toxin-resistant population of *D. yakuba* on Mayotte becoming a separate species to the mainland population of *D. yakuba*.

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- (c) Suggest how scientists can use bioinformatics to estimate the length of time that the population of *D. yakuba* has been living on Mayotte.

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